

**Directions.** Complete all the problems. Your solutions to the problems should be submitted to Gradescope before the indicated due date above. Please follow rules regarding Gradescope submission as described in the syllabus. If you use any outside source in your solutions, you must cite the source.

**Problem 1.** Consider the IVP

$$y' = 2x(1 + y), \quad y(0) = 0$$

- (a) Find the solution to the IVP.
- (b) Starting with  $y_0(t) \equiv 0$ , find the Picard iterates  $y_1(t), y_2(t), y_3(t), y_4(t)$ .

**Solution.** (a) We solve the IVP

$$y' = 2x(1 + y), \quad y(0) = 0.$$

Rewrite as

$$y' - 2xy = 2x.$$

The integrating factor is

$$\mu(x) = e^{-x^2}.$$

Multiplying the equation by  $\mu$  gives

$$(ye^{-x^2})' = 2xe^{-x^2}.$$

Integrating from 0 to  $t$ ,

$$y(t)e^{-t^2} - y(0) = \int_0^t 2se^{-s^2} ds.$$

Using  $y(0) = 0$  and

$$\int 2se^{-s^2} ds = -e^{-s^2} + C,$$

we obtain

$$y(t)e^{-t^2} = -e^{-t^2} + C.$$

Thus

$$y(t) = -1 + Ce^{t^2}.$$

Applying the initial condition  $y(0) = 0$  shows  $C = 1$ , so the solution is

$$\boxed{y(t) = e^{t^2} - 1}.$$

(b) Picard iteration is defined by

$$y_{n+1}(t) = y(0) + \int_0^t 2s(1 + y_n(s)) ds, \quad y_0(t) \equiv 0.$$

$$y_1(t) = \int_0^t 2s ds = t^2,$$

$$y_2(t) = \int_0^t 2s(1 + s^2) ds = t^2 + \frac{t^4}{2},$$

$$y_3(t) = \int_0^t 2s \left( 1 + s^2 + \frac{s^4}{2} \right) ds = t^2 + \frac{t^4}{2} + \frac{t^6}{6},$$

$$y_4(t) = \int_0^t 2s \left( 1 + s^2 + \frac{s^4}{2} + \frac{s^6}{6} \right) ds = t^2 + \frac{t^4}{2} + \frac{t^6}{6} + \frac{t^8}{24}.$$

In general,

$$y_n(t) = \sum_{k=1}^n \frac{t^{2k}}{k!},$$

and  $y_n(t) \rightarrow e^{t^2} - 1$ , matching the exact solution.

**Problem 2.** Consider the IVP

$$y'(t) = y(t), \quad y(0) = 1$$

In lecture, we found the first several Picard iterates, starting (as usual) with  $y_0(t) \equiv y(0) = 1$ . In this problem, we consider what happens if we start with a different initial function  $y_0(t)$ . Find the Picard iterates  $y_1(t), y_2(t), y_3(t)$  with the following **different choices** of initial Picard iterate.

(a)  $y_0(t) = e^t$

(b)  $y_0(t) = \cos t$

(c)  $y_0(t) = (t + 1)^2$

**Solution.** Picard iteration for the IVP  $y' = y$ ,  $y(0) = 1$  is

$$y_{n+1}(t) = 1 + \int_0^t y_n(s) ds,$$

and we are given different choices of  $y_0(t)$ .

(a)  $y_0(t) = e^t$ . Then

$$y_1(t) = 1 + \int_0^t e^s ds = 1 + (e^t - 1) = e^t,$$

so  $y_1(t) = e^t$ . Hence all subsequent iterates are the same:

$$\boxed{y_1(t) = y_2(t) = y_3(t) = e^t.}$$

(b)  $y_0(t) = \cos t$ . Then

$$y_1(t) = 1 + \int_0^t \cos s ds = 1 + \sin t,$$

$$y_2(t) = 1 + \int_0^t (1 + \sin s) ds = 1 + (t - \cos t + 1) = 2 + t - \cos t,$$

$$y_3(t) = 1 + \int_0^t (2 + s - \cos s) ds = 1 + (2t + \frac{t^2}{2} - \sin t).$$

Thus

$$\boxed{y_1(t) = 1 + \sin t, \quad y_2(t) = 2 + t - \cos t, \quad y_3(t) = 1 + 2t + \frac{t^2}{2} - \sin t.}$$

(c)  $y_0(t) = (t + 1)^2 = t^2 + 2t + 1$ . Compute

$$y_1(t) = 1 + \int_0^t (s + 1)^2 ds = 1 + \left( \frac{t^3}{3} + t^2 + t \right),$$

$$y_2(t) = 1 + \int_0^t \left( 1 + s + s^2 + \frac{s^3}{3} \right) ds = 1 + \left( t + \frac{t^2}{2} + \frac{t^3}{3} + \frac{t^4}{12} \right),$$

$$y_3(t) = 1 + \int_0^t \left( 1 + s + \frac{s^2}{2} + \frac{s^3}{3} + \frac{s^4}{12} \right) ds = 1 + \left( t + \frac{t^2}{2} + \frac{t^3}{6} + \frac{t^4}{12} + \frac{t^5}{60} \right).$$

So

$$\boxed{y_1(t) = 1 + t + t^2 + \frac{t^3}{3}, \quad y_2(t) = 1 + t + \frac{t^2}{2} + \frac{t^3}{3} + \frac{t^4}{12}, \quad y_3(t) = 1 + t + \frac{t^2}{2} + \frac{t^3}{6} + \frac{t^4}{12} + \frac{t^5}{60}.}$$

(As expected, the iterates approach the true solution  $e^t$ .)

**Problem 3.** Let  $f : [0, a] \times \mathbb{R} \rightarrow \mathbb{R}$  and assume  $f = f(t, y)$  is continuous on its domain and Lipschitz in  $y$ , with Lipschitz constant  $L \geq 0$ . Consider the IVP

$$y' = f(t, y), \quad y(0) = y_0,$$

and let  $y_0, y_1, y_2, \dots$  be the Picard iterates. Suppose  $\bar{y}$  is a solution to the IVP, and let  $\bar{M} = \|\bar{y} - y_0\|_\infty$ . Prove by induction on  $n$  that

$$|\bar{y}(t) - y_n(t)| \leq \frac{\bar{M}L^n t^n}{n!}$$

for all  $t \in [0, a]$ .

**Solution.** Let  $\bar{y}$  be a solution of the IVP and write  $\bar{M} = \|\bar{y} - y_0\|_\infty = \sup_{t \in [0, a]} |\bar{y}(t) - y_0(t)|$ . We prove by induction on  $n \geq 0$  that

$$|\bar{y}(t) - y_n(t)| \leq \frac{\bar{M}L^n t^n}{n!}, \quad t \in [0, a].$$

**Base case  $n = 0$ .** By definition of  $\bar{M}$ ,

$$|\bar{y}(t) - y_0(t)| \leq \bar{M} = \frac{\bar{M}L^0 t^0}{0!},$$

so the claim holds for  $n = 0$ .

**Inductive step.** Suppose the inequality holds for some  $n \geq 0$ . Using the integral form of the Picard iterate,

$$y_{n+1}(t) = y_0 + \int_0^t f(s, y_n(s)) ds, \quad \bar{y}(t) = y_0 + \int_0^t f(s, \bar{y}(s)) ds,$$

so

$$\bar{y}(t) - y_{n+1}(t) = \int_0^t [f(s, \bar{y}(s)) - f(s, y_n(s))] ds.$$

Apply the Lipschitz property in the  $y$ -variable (constant  $L$ ) to get

$$|\bar{y}(t) - y_{n+1}(t)| \leq \int_0^t L |\bar{y}(s) - y_n(s)| ds.$$

By the induction hypothesis,

$$|\bar{y}(s) - y_n(s)| \leq \frac{\bar{M}L^n s^n}{n!} \quad \text{for } 0 \leq s \leq t,$$

hence

$$|\bar{y}(t) - y_{n+1}(t)| \leq \int_0^t L \cdot \frac{\overline{ML}^n s^n}{n!} ds = \frac{\overline{ML}^{n+1}}{n!} \int_0^t s^n ds = \frac{\overline{ML}^{n+1} t^{n+1}}{(n+1)!}.$$

This is exactly the desired estimate for  $n+1$ . By induction the inequality holds for all  $n \geq 0$  and all  $t \in [0, a]$ .

**Problem 4.** Suppose  $f, f_1, f_2, \dots$  are all integrable real-valued functions on  $[0, 1]$ . Prove that if  $\|f_n - f\|_\infty \rightarrow 0$  as  $n \rightarrow \infty$ , then

$$\lim_{n \rightarrow \infty} \int_0^1 f_n(t) dt = \int_0^1 f(t) dt.$$

**Solution.** Since  $\|f_n - f\|_\infty \rightarrow 0$  as  $n \rightarrow \infty$ , for every  $\varepsilon > 0$  there exists  $N$  such that for all  $n \geq N$  and all  $t \in [0, 1]$ ,

$$|f_n(t) - f(t)| \leq \|f_n - f\|_\infty < \varepsilon.$$

Using integrability and the triangle inequality,

$$\left| \int_0^1 f_n(t) dt - \int_0^1 f(t) dt \right| = \left| \int_0^1 (f_n(t) - f(t)) dt \right| \leq \int_0^1 |f_n(t) - f(t)| dt.$$

For  $n \geq N$  the integrand is bounded by  $\varepsilon$ , so

$$\int_0^1 |f_n(t) - f(t)| dt \leq \int_0^1 \varepsilon dt = \varepsilon.$$

Thus for all  $n \geq N$ ,

$$\left| \int_0^1 f_n - \int_0^1 f \right| \leq \varepsilon,$$

which shows

$$\lim_{n \rightarrow \infty} \int_0^1 f_n(t) dt = \int_0^1 f(t) dt.$$

**Problem 5.** Show that  $y(t) = 0$  and  $y(t) = t^3$  are both solutions of the initial value problem

$$y' = 3y^{2/3}, \quad y(0) = 0$$

Explain why this fact does not contradict the Picard-Lindelof theorem. (Put a slightly different way, what does Picard-Lindelof empower you to say about the function  $f(t, y) = 3y^{2/3}$ ?)

**Solution.** Showing that they are solutions should be straightforward. This fact does not contradict the Picard-Lindelof theorem because there is no interval containing  $(0, 0)$  for which  $f(t, y)$  is defined for  $f_y(t, y)$ . Thus there is no interval for which  $f(t, y)$  can be Lipschitz in  $y$ .

**Problem 6.** Consider the differential equation

$$y' = \frac{1}{y+1}.$$

For what initial conditions are we guaranteed that a unique solution exists on some interval around the initial time?

**Solution.** The right-hand side  $f(y) = \frac{1}{y+1}$  is continuous (and  $C^1$ ) for all  $y \neq -1$ . In particular, if  $y_0 \neq -1$  then there exists a rectangle

$$R = \{(t, y) : |t - t_0| \leq h, |y - y_0| \leq r\}$$

that does not meet the line  $y = -1$ . On  $R$  the function  $f(t, y) = 1/(y + 1)$  is continuous in  $t, y$  and Lipschitz in  $y$  (indeed  $f_y(y) = -1/(y + 1)^2$  is continuous and bounded on  $R$ ), so the Picard–Lindelöf theorem gives existence and uniqueness of a solution of the IVP on some interval around  $t_0$ .

If  $y_0 = -1$  the right-hand side is not defined at the initial value, so the standard existence theorems do not apply. There is no (classical) solution passing through the point  $(t_0, -1)$  because any differentiable function  $y$  with  $y(t_0) = -1$  would have to satisfy  $y'(t_0) = 1/(y(t_0) + 1) = 1/0$ , which is impossible.

For additional insight one can separate variables and obtain the explicit general solution (on intervals where  $y \neq -1$ ):

$$(y + 1) dy = dt \implies \frac{1}{2}(y + 1)^2 = t + C,$$

so

$$y(t) = -1 \pm \sqrt{2(t + C)}.$$

These explicit solutions are defined only where the square root is real; they never cross the line  $y = -1$  (they reach  $y = -1$  exactly where the square root vanishes), which confirms that no solution can pass through  $y = -1$ .

*Conclusion:* A unique solution exists on some interval around the initial time precisely for initial conditions  $y(t_0) = y_0$  with  $y_0 \neq -1$ . For  $y_0 = -1$  the IVP has no classical solution.

**Problem 7.** Consider the following initial value problem

$$ty' = y + 2t, \quad y(t_0) = 3.$$

Answer the following Without solving the initial value problem.

- (a) Determine all values  $t_0$  for which we are guaranteed a unique solution exists on some interval around  $t_0$ . Justify your answer.
- (b) For the values of  $t_0$  obtained in part (a), determine for which intervals of time  $[c, d]$ ,  $c \leq t_0 \leq d$ , we can guarantee there is a unique solution with domain  $[c, d]$ .

**Solution.** Rewrite the equation as

$$y' = \frac{1}{t}y + 2, \quad t \neq 0.$$

Thus the right-hand side can be viewed as the function

$$f(t, y) = \frac{1}{t}y + 2.$$

(a) The Picard–Lindelöf (existence and uniqueness) theorem requires  $f$  to be continuous in  $t$  and Lipschitz in  $y$  on a domain containing the initial point  $(t_0, 3)$ . For  $t \neq 0$  the function  $f(t, y)$  is continuous in  $t$  and  $y$ , and

$$\frac{\partial f}{\partial y}(t, y) = \frac{1}{t}$$

is continuous for  $t \neq 0$ ; hence  $f$  is Lipschitz in  $y$  on any interval that does not meet the line  $t = 0$ . Therefore the hypotheses of Picard–Lindelöf hold whenever  $t_0 \neq 0$ , and we are guaranteed a unique solution on some interval about  $t_0$ .

If  $t_0 = 0$  the right-hand side is not defined at  $t = 0$ , so the standard existence–uniqueness theorem does not apply; no guarantee of a unique (classical) solution through  $(0, 3)$  can be made by the knowledge we have in this course.

(b) Fix  $t_0 \neq 0$ . To guarantee a unique solution on a closed interval  $[c, d]$  containing  $t_0$  we must choose  $[c, d]$  so that  $f$  is continuous and Lipschitz in  $y$  on a rectangle containing  $[c, d] \times \{3\}$ . Equivalently,  $[c, d]$  must not contain 0. Thus any closed interval  $[c, d]$  with either

$$0 < c \leq t_0 \leq d \quad (\text{i.e. } c > 0)$$

or

$$c \leq t_0 \leq d < 0 \quad (\text{i.e. } d < 0)$$

will do. On such an interval  $[c, d]$  the quantity  $\sup_{t \in [c, d]} |1/t|$  is finite, so  $f$  is Lipschitz in  $y$  there and Picard–Lindelöf guarantees a unique solution on  $[c, d]$ .